

Lecture 5: Rationalizable and Nash Equilibrium

FEBRUARY 2

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5.1 Equilibria in Games

5.1.1 Rationalizable Equilibrium

Now that we have defined belief, we can start to discuss what happens when a rational person has a belief.

Definition 5.1 (rationalizable strategy) *A rationalizable strategy is a strategy that remains after we eliminate any strategy that is not a best response to the player's beliefs.*

		Player 2	
		<i>A</i>	<i>B</i>
Player 1	<i>A</i>	2, 1	0, 0
	<i>B</i>	0, 0	1, 2

Example 5.2 *We see that $BR_1(A) = \{A\}$, $BR_1(B) = \{B\}$, so we cannot eliminate any strategy from player 1's strategy set. This is since both actions could be the ideal move.*

This means the set of rationalizable strategies are $\{(A, A), (A, B), (B, A), (B, B)\}$.

Example 5.3 (Left, Right, Center) *Let's play this game: $N = \{\text{Player 1, Player 2}\}$, $S_1 = \{U, M, D\}$, $S_2 = \{L, C, R\}$ with the following payoff matrix. Recall that $\{(U, L)\}$ was the IESDS equilibrium*

		Player 2		
		<i>L</i>	<i>C</i>	<i>R</i>
Player 1	<i>U</i>	4, 3	5, 1	6, 2
	<i>M</i>	2, 1	8, 4	3, 6
	<i>D</i>	3, 0	9, 6	2, 8

*We see that $BR_1(L) = \{U\}$, $BR_1(C) = \{D\}$, $BR_1(R) = \{U\}$. This means that *M* is not a best response to any belief player 1 can have. So *M* is not rationalizable.*

*Now we look at $BR_2(U) = \{L\}$, $BR_2(D) = \{R\}$, which means that *C* is never rationalizable.*

*Now we look at $BR_1(L) = \{U\}$, $BR_1(R) = \{U\}$, so *D* is not rationalizable and *U* is a dominating strategy.*

Lastly, $BR_2(U) = \{L\}$, so the set of rationalizable outcomes are $\{(U, L)\}$.

Remark 5.4 We can eliminate M because it isn't rationalizable, but when we were doing IESDS last time, we couldn't eliminate M since neither U or D strictly dominated it. This means that IESDS is a broader category of equilibrium. But are they the same?

		Player 2	
		L	R
Player 1	U	3,0	0,0
	M	0,0	3,0
	D	1,0	1,0

Example 5.5 Here we see that there are no strictly dominated strategies for either player. This means that the set of IESDS-equilibria are $\{(U, L), (M, L), (D, L), (U, R), (M, R), (D, R)\}$.

However, we have seen that $BR_1(L) = \{U\}$, $BR_1(R) = \{M\}$, so D isn't rationalizable. However, we cannot simplify any more so the set of all rationalizable equilibria are $\{(U, L), (M, L), (U, R), (M, R)\}$.

5.1.2 Nash Equilibria

Definition 5.6 A Nash Equilibrium is when all players are playing a best response against each other.

		Player 2	
		A	B
Player 1	A	2,1	0,0
	B	0,0	1,2

Example 5.7 We can see that $BR_1(A) = \{A\}$, $BR_2(A) = \{A\}$. Since both players are playing their best response against each other, then (A, A) is a Nash equilibrium. Additionally, $BR_1(B) = \{B\}$, $BR_2(B) = \{B\}$, which means (B, B) is also a Nash equilibrium.

Example 5.8 Let's look at this new problem that is the same structure of the Left, Right, Center game, but with different payoffs.

		Player 2		
		L	C	R
Player 1	U	7, 7	4, 2	1, 8
	M	2, 4	5, 5	2, 3
	D	8, 1	3, 2	0, 0

We could first start with the IESDS, strategy where R is dominated by L and D is dominated by U . But after eliminating these, we see that we cannot go further.

If we start with rationalizable strategies, we see

$$\begin{aligned} BR_1(L) &= \{D\} & BR_2(U) &= \{R\} \\ BR_1(C) &= \{M\} & BR_2(M) &= \{C\} \\ BR_1(R) &= \{M\} & BR_2(D) &= \{C\} \end{aligned}$$

We could reduce the problem to find the rationalizable equilibrium, but we already see that the Nash equilibrium is $\{(M, C)\}$ because $BR_1(C) = \{M\}$, $BR_2(M) = \{C\}$, i.e. the two players are playing their best responses against each other. This is visualized in the table below (Tab: 5.8)

		Player 2		
		<i>L</i>	<i>C</i>	<i>R</i>
Player 1	<i>U</i>	7, 7	4, 2	1, 8
	<i>M</i>	2, 4	<u>5, 5</u>	<u>2, 3</u>
	<i>D</i>	<u>8, 1</u>	3, 2	0, 0

		Other Farmer				
		1	2	3	4	5
You	1	2.3026	2.1972	2.0794	1.9459	1.7918
	2	2.8904	2.7726	2.6391	2.4849	2.3026
	3	3.1781	3.0445	2.8904	2.7081	2.4849
	4	3.3322	3.1781	2.9957	2.7726	2.4850
	5	3.4012	3.2189	2.9958	2.7082	2.3026

Table 5.1: Tragedy of the Commons represented by the trade off between owning cows and having a beautiful pasture

Example 5.9 (Tragedy of the commons) *Imagine that there is a beautiful pasture that you and your whole ranching community can enjoy. There is also so much room for your cattle to graze. The question is: how many cattle should you get? How many should your neighbors get? You obviously get more utility from having more cows as you can sell more milk, but the cows need grazing area, which reduces the satisfaction of the beautiful pasture.*

While we can think of n players, this is very hard to compute in practice. Therefore we will only consider two farmers and up to 5 cows. Note each farmer must have a positive number of cows to feed their family. This is represented in the table below in Table 5.9.

Since this game is symmetric, and both players are rational, it is simpler to consider what you would do, and know that the other rancher would do the same.

We see that no matter how many cows your neighbor has, we get the most payoff from always having 4 or 5 cows. This means both you and your neighbor wouldn't consider owning 1-3 cows. This is seen in Table 5.9.

We now see that owning 4 cows strictly dominates owning 5 cows. So both you and your neighbor will own 4 cows!

However, if we want to maximize total payoff, which is the sum of the payoff functions, we see represented in Table 5.9. The largest entry in this table is the option that maximizes total utility, which we see happens when both farmers own 3 cows.

This means the difference between what will happen (Nash Equilibrium) when both players are trying to maximize their own profit is lower than the socially optimal case (maximizing total utility). The difference is $5.5452 - 5.7807 = -0.2355$.

Remark 5.10 *This is a conflict among scarce resources, where there is tension between the individual's selfish nature and the common good. The "tragedy" is that behaving selfishly is detrimental to society.*

		Other Farmer	
		4	5
You	4	2.7726	2.4850
	5	2.7082	2.3026

		Other Farmer				
		1	2	3	4	5
You	1	4.6052	5.0876	5.2575	5.2781	5.1930
	2	5.0876	5.5452	5.6836	5.6630	5.5215
	3	5.2575	5.6836	5.7807	5.7038	5.4806
	4	5.2781	5.6630	5.7038	5.5452	5.1930
	5	5.1930	5.5215	5.4806	5.1930	4.6052

Table 5.2: Total Utility of owning different number of cows.

This leads us nicely into Parato Optimal solutions. We look at cases where what will happen isn't optimal in some sense. We have seen it many times before the Tragedy of the Commons.

Definition 5.11 *A strategy profile Pareto dominates another strategy if it is at least as good for all possible payoffs, and strictly better than the other strategy for at least one strategy. If a strategy profile isn't Pareto dominated, then it is Pareto optimal.*

Example 5.12 (Pareto Optimal strategies for Prisoner's Dilemma) *Recall the Prisoner's dilemma:*

		Player 2	
		C	S
Player 1	C	-4,-4	-1,-5
	S	-5,-1	-2,-2

The prisoners will always end up confessing and both achieving 4 years of jail time. However, this is not Pareto Optimal. First we see that (S, S) is better for both of the prisoners. Since it is at least as good (actually strictly better) for each player, (S, S) Pareto dominates (C, C) .

Now let's look at if player 1 chooses to stay silent and player 2 chooses to confess. Player 1's payoff is lower than the payoff to stay silent ($-5 < -2$), but since player 2 is better off to confess in this situation ($-1 > -2$), this strategy profile is NOT Pareto dominated by (S, S) . This means the set of Pareto optimal profiles are $\{(S, C), (C, S), (S, S)\}$.

Remark 5.13 *Don't confuse Pareto optimality with the best "symmetric" outcome that leaves all players "equally" happy.*